

Framework of Live Doubt Session

- Session starts at 5:15 PM
- Weekly recap of topics (via PPT)
- Students may post doubts in the chat during the presentation
- Topic-wise doubt clarification after the recap
- Assignment-related doubts addressed at the end

Recap for Week 3

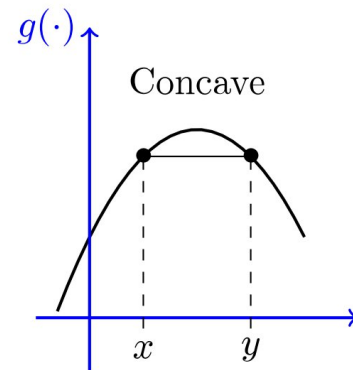
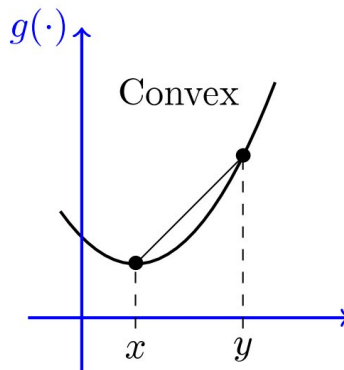
- Convex vs non convex functions.
- First Order and Second Order Characterization of Convexity
- Positive definite and semi-definite matrices.
- Hessian Shapes
- Role of Convexity in optimisation.
- Numerical schemes for unconstrained optimisation.
- Line search
- Gradient Descent Method with example.
- Univariate Unconstrained , Multivariate Unconstrained Optimisations
- Fermat's Rule
- Hessian Sufficiency (Diff from Hessian Matrix)

Convex vs Non-Convex Functions :-

Geometric Meaning of Convex - Only **one global minimum**

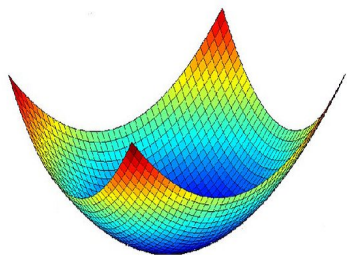
Real Life Example - Cost minimization in manufacturing ,

Linear regression loss function (MSE)

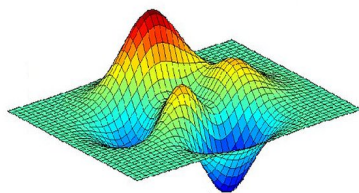


Non-Convex Function - Multiple local minima

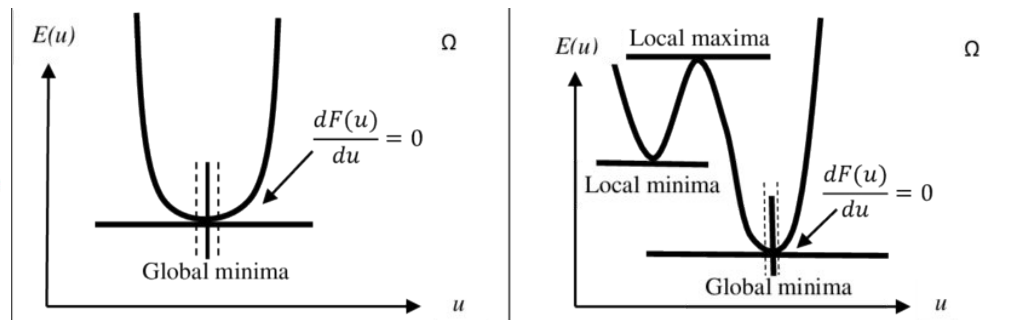
Hard to find global minimum , Neural networks (deep learning) are non-convex



convex function



non-convex function



Real Life Example - Deep neural network training , Portfolio optimization with complex constraints

First Order Characterization of Convexity: - $f(y) \geq f(x) + \nabla f(x)^T (y-x)$

Meaning: Function always lies **above its tangent plane**.

Intuition: Gradient gives a supporting hyperplane.

Second Order Characterization of Convexity: - For twice differentiable function: $\nabla^2 f(x) \succeq 0$, $\rightarrow$ Strictly convex.

(Hessian matrix is positive semi-definite (PSD))

Positive Definite & Semi-Definite Matrices :-

Positive Definite (PD): - $X^T A X > 0$, X = Matrix , X^T = Transpose of Matrix

All eigenvalues > 0 , Surface shaped like bowl , Unique minimum

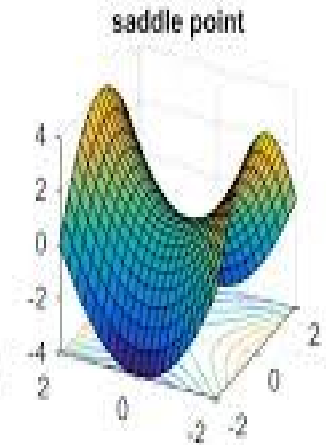
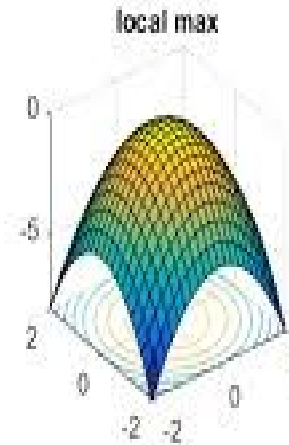
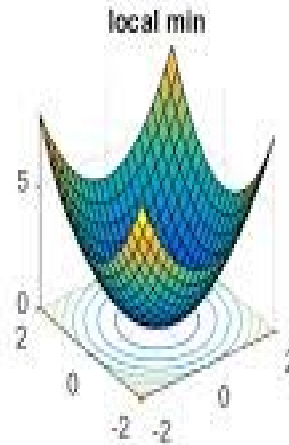
Positive Semi-Definite (PSD) :- $X^T A X \geq 0$, X = Matrix , X^T = Transpose of Matrix

Eigenvalues ≥ 0 , Flat directions possible

Real Example :- Covariance matrix in ML $\rightarrow$ Always PSD.

Hessian Shapes:

Hessian Type	Shape	Conclusion
PD (Positive Definite)	Bowl	Local Minima
ND (Non Definite)	Upside Bowl	Local Maxima
Indefinite	Saddle	Saddle Point



Fermat's Rule: If x^* is a local minimum and differentiable :- $\nabla f(x^*) = 0$, $x^* \rightarrow$ Optimal solution

Important Condition : Gradient zero is **necessary**, not sufficient.

Hessian Sufficiency (How Hessian Sufficiency is Different from Hessian Matrix ?)

Hessian Matrix = Matrix of second derivatives.

Hessian Sufficiency = Condition using Hessian to classify stationary point.

If , $\nabla f(x^*)=0$

Hessian PD $\rightarrow$ Local minimum

Hessian ND $\rightarrow$ Local maximum

Hessian indefinite $\rightarrow$ Saddle

Unconstrained Optimization:-

Univariate (Single Variable)

Example: $f(x) = x^2 - 4x + 3$; Step 1: Derivative : $f'(x) = 2x - 4$; Step 2: Set zero : $2x - 4 = 0$, $x = 2$

Step 3: Second derivative : $f''(x) = 2 > 0$, Minimum at $x = 2$

Multivariate

Example: $f(x,y) = x^2 + y^2$; Gradient: $\nabla f = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$; Set zero : $x=0$, $y=0$; Hessian: $H = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$; PD $\rightarrow$ Global minimum

Numerical Schemes for Unconstrained Optimization

When analytical solution not possible: 1) Gradient Descent $x_{k+1} = x_k - \alpha \nabla f(x_k)$ 2) Newton's Method $x_{k+1} = x_k - H^{-1} \nabla f$

3) Quasi-Newton (BFGS) - Used in ML libraries.

Line Search We choose step size α .

Two types:

- Exact Line Search
- Backtracking Line Search

Goal: Minimize $f(x_k - \alpha \nabla f)$

Gradient Descent Example

Let: $f(x)=x^2$; Derivative: $f'(x)= 2x$; Update: $x_{k+1}=x_k-\alpha 2x_k$; If $\alpha=0.1$;

Start $x_0=5$,

$$x_1= 5 - 0.1(10) = 4$$

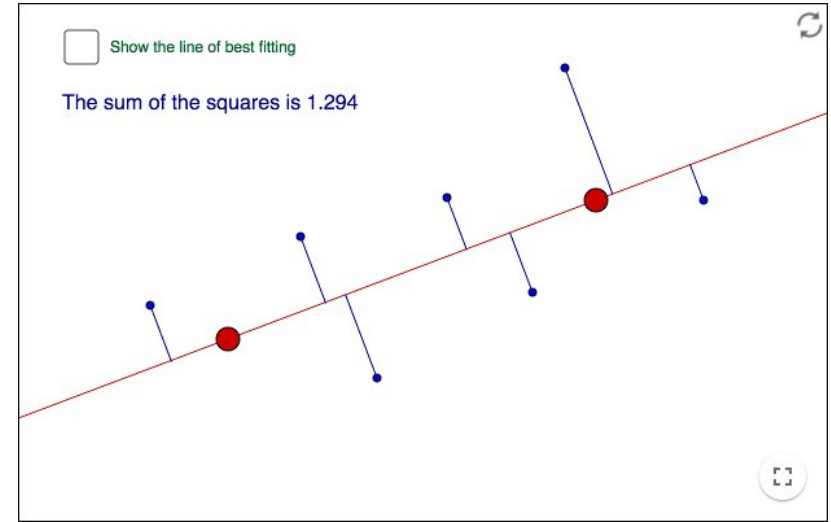
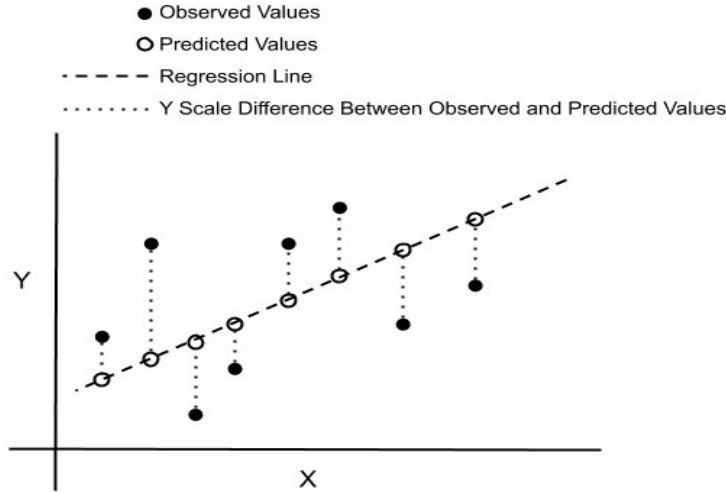
$$x_2=4 - 0.1(8) = 3.2 \text{ , Converges toward zero.}$$

Week 4

- Least square Method.
- Step Length
- Lipschitz function and Lipschitz Gradient
- Exact and Inexact Line search
- Inexact Line Search- Armijo and Wolfe's Conditions
- Zoutendijk's Theorem
- Backtracking

Least Squares Method - Least Squares finds parameters that **minimize the squared error** between predicted values and real data.

$$\min \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad \text{If model is linear : } y = w_0 + w_1 x. \text{ Then } w_0, w_1 \text{ so the total squared error is minimum}$$



Real Life Example -

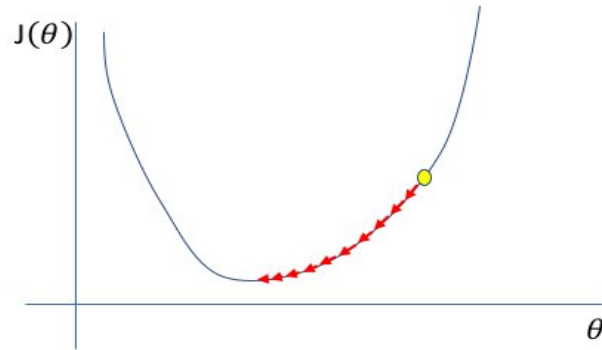
Area	Actual Price
800	50K
900	55K
1000	65K

Model : Price = $w_0 + w_1 \times \text{Area}$, Least squares finds **best line**.

AI / ML usage : Linear Regression , Polynomial Regression , Kalman Filters

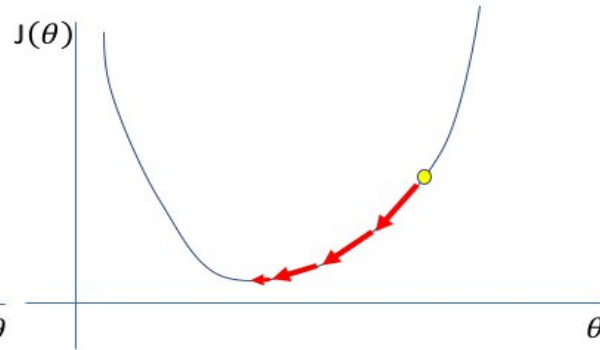
Step Length (Learning Rate) - In gradient descent: $x_{k+1} = x_k - \alpha \nabla f(x_k)$, α - step length / learning rate

Too low



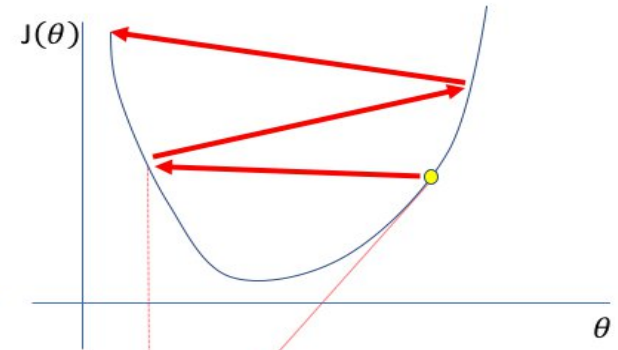
A small learning rate requires many updates before reaching the minimum point

Just right



The optimal learning rate swiftly reaches the minimum point

Too high



Too large of a learning rate causes drastic updates which lead to divergent behaviors

If step length is :

Step Size	Result
Too Small	Slow Learning
Too Big	Oscillations
Proper	Fast Convergence

Real Life Example - Walking downhill ,

Big steps $\rightarrow$ may slip past valley , Small steps $\rightarrow$ reach slowly

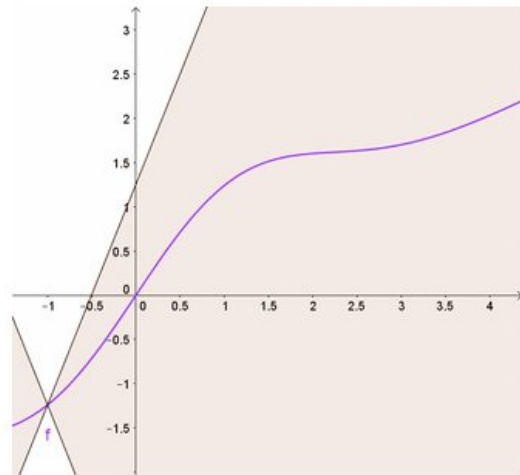
AI / ML - Just keep changing α - 0.1, 0.01 , 0.001 etc

Lipschitz Function - A function is **Lipschitz continuous** if its slope cannot change too fast. $|f(x_1) - f(x_2)| \leq L |x_1 - x_2|$, L = Lipschitz

In particular, a **real-valued function** $f: \mathbb{R} \rightarrow \mathbb{R}$ is called Lipschitz continuous if there exists a positive real constant K such that, for all real x_1 and x_2 ,

$$|f(x_1) - f(x_2)| \leq K |x_1 - x_2|.$$

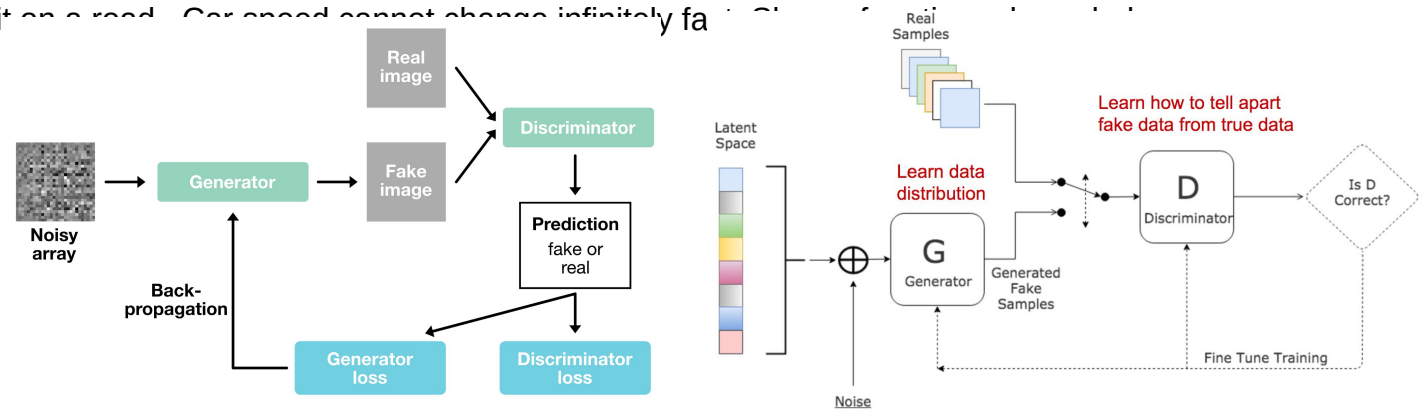
For a Lipschitz continuous function, there exists a double cone (white) whose origin can be moved along the graph so that the whole graph always stays outside the double cone



Real Life Example : - Speed limit on a road. Car speed cannot change infinitely fast.

AI / ML Example :-

GAN training stability,
Wasserstein GAN
(Generative Adversarial Networks)



Lipschitz Gradient - A function has Lipschitz gradient if $\| \nabla f(x) - \nabla f(y) \| \leq L \|x - y\|$,

Mathematical Meaning:- Gradient changes smoothly.

Why it's Important ? Reason - Ensures **gradient descent converges**.

ML Example - Loss functions like: Mean Squared Error , Logistic Loss. (have Lipschitz gradients).

Exact Line Search - We find step size that **exactly minimizes** function along search direction. $\alpha = \arg \min f(x_k + \alpha d_k)$.

Real Life Example - Finding lowest point **exactly along a straight road**.

ML Example - Rarely used because of expensive , large datasets

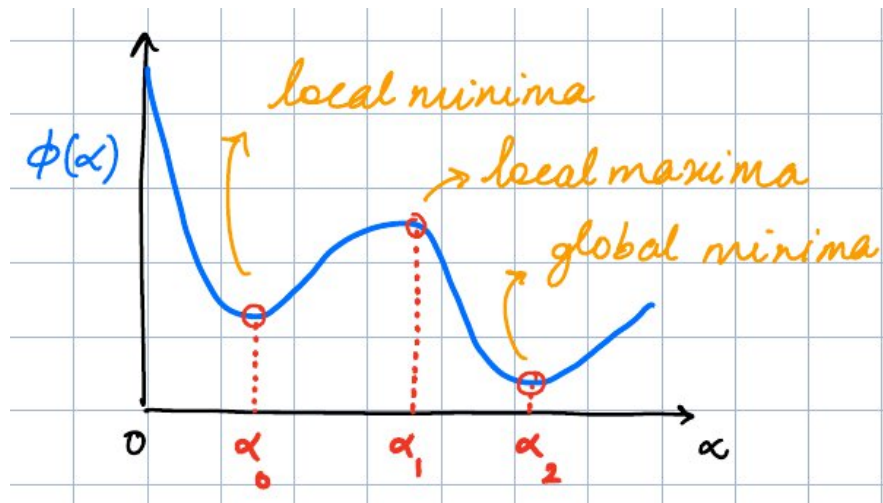
Inexact Line Search

Instead of exact minimum, choose **good enough step size**.

Faster for real problems.

Used in:- Gradient descent , Quasi-Newton methods ,

Deep learning optimizers



Armijo Condition : Ensures **sufficient decrease** in function, $f(\mathbf{x}_k + \alpha \mathbf{d}_k) \leq f(\mathbf{x}_k) + c_1 \alpha \nabla f(\mathbf{x}_k)^T \mathbf{d}_k$.

Mathematical Meaning : Step must reduce loss sufficiently

Real Life Example - Negotiation discount (we accept **reasonable discount**, not perfect one)

ML Example - Used in : Backtracking line search , Large scale optimization

Wolfe Conditions

Two conditions - 1) Armijo condition - Decrease condition , 2) Curvature condition : $\nabla f(\mathbf{x}_k + \alpha \mathbf{d}_k)^T \mathbf{d}_k \geq c_2 \nabla f(\mathbf{x}_k)^T \mathbf{d}_k$.

Ensures slope is reduced

Why it's Important ? Reason - Guarantees convergence

ML Example :- Used in: BFGS optimizer , L-BFGS

Zoutendijk's Theorem

It states: if 1) descent direction used and 2) Wolfe conditions satisfied

Then $\sum \cos^2(\theta_k) \|\nabla f(\mathbf{x}_k)\|^2 < \infty$, **Mathematical meaning** - Gradient approaches zero (So algorithm converges).

Real Life Meaning- If we keep moving **downhill properly**, we will eventually reach valley.

Backtracking Line Search

Simple algorithm to choose step size.

Start with : $\alpha = 1$, Then reduce: $\alpha = \beta\alpha$, until Armijo condition satisfied.

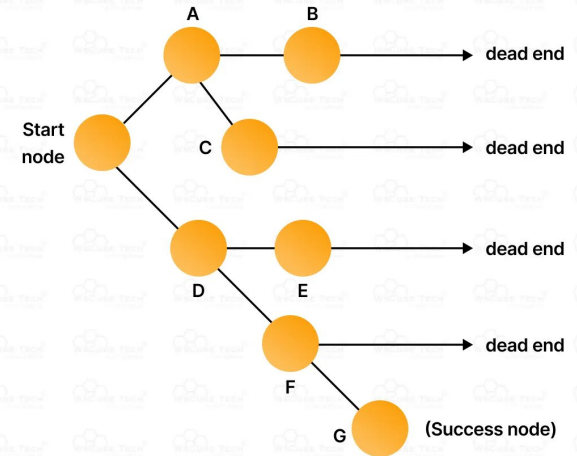
Real Life Example - We trying to park car

If car goes too far: reverse slightly , adjust steps.

ML Example :- Backtracking used in : Newton optimization ,

Deep learning training

Backtracking Algorithm



Week 5

- Table of Gradient Descent Variants: SGD, Mini-batch GD, Momentum, AdaGrad, RMSProp, Adam
- Pure Newton's Method and Damped Newton Method
- Quasi Newton Method and its methods:
 - Davidon-Fletcher powel Method
 - BFGS Method

Gradient Descent Variants - Gradient descent minimizes a loss function : $\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)$

Where $\mathbf{x}_k$ = current parameters , α = learning rate , $\nabla f(\mathbf{x})$ = gradient

Method	Idea	Real life Analogy	AI/ML Usage
SGD	Use one Data Sample at a time	Learning from 1 Mistake at a time	Online Learning
Mini - Batch GD	Use small batch of data	Studying few Problems at once	Deep Learning
Momentum	Uses past direction to accelerate	Rolling Ball gaining speed	CNN Training
AdaGrad	Learning rate adjusts per parameter	Slowing down after many steps	NLP Sparse Features
RMS Prop	Fixes AdaGrad Decay issue	Adaptive walking speed	RNN Training
Adam	Combines Momentum + RMSProp	Smart driver adjusting speed & direction	Most Common Optimiser

SGD (Stochastic Gradient Descent) : $\theta = \theta - \alpha \nabla f_i(\theta)$, Uses **one data point at a time**.

Real-Life Example :- Learning to play cricket. (we adjust our technique after each ball.)

AI/ML Example - Neural networks , Online recommendation systems , Reinforcement learning

Mini-Batch Gradient Descent : Uses small batch of data. Example a Batch size = 16 /18 /20 /26 /30.

Real-Life Example - Instead of correcting after every exam question, you analyze 20 questions together.

AI/ML Example - Most deep learning training uses: 16 , 32 ,64,128

Momentum :- Update rule: $\mathbf{v}_t = \beta \mathbf{v}_{t-1} + \alpha \nabla f(\mathbf{x})$, $\mathbf{x}_{t+1} = \mathbf{x}_t - \mathbf{v}_t$ (Uses **past gradients** to accelerate.)

Real-Life Example - Rolling a **heavy ball downhill**. (Momentum keeps it moving.)

GRADIENT DESCENT



OPTIMIZATION
PATH



LOSS FUNCTION
MINIMIZATION



ITERATIVE
REFINEMENT



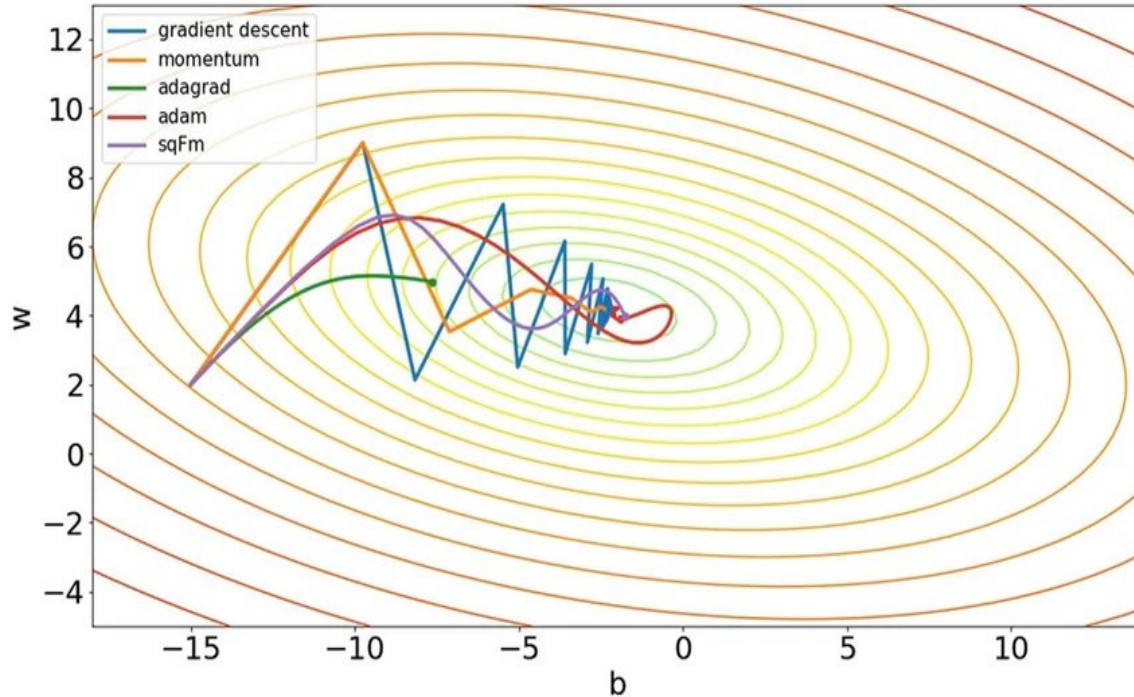
CONVERGENCE



GRADIENT
MOVEMENT

AI/ML Example - Used in training :

CNN models , computer vision networks



AdaGrad : Adapts learning rate per parameter. $\theta = \theta - \frac{\alpha}{\sqrt{G_t + \epsilon}} \nabla f$ Where G_t is sum of squared gradients.

Real-Life Example - When learning piano (we practice **difficult keys more carefully**.)

AI/ML Example - Sparse datasets , NLP tasks

RMSProp : Fixes AdaGrad problem (learning rate decay). $E[g^2]_t = \beta E[g^2]_{t-1} + (1-\beta) g_t^2$

Real-Life Example - Walking on terrain while adjusting step length dynamically.

AI/ML Example : RNN training , speech recognition models

Adam (Adaptive Moment Estimation) : It Combines - Momentum + RMSProp

Real-Life Example -

Smart driver (remembers past speed + adjusts according to road conditions)

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \end{aligned}$$

AI/ML Example - Most common optimizer : transformers , GPT models , computer vision , reinforcement learning

Newton's Method : Uses **second derivative (Hessian)**. $x_{k+1} = x_k - H^{-1} \nabla f$, where H = Hessian matrix.

Real-Life Example - Using **curvature of road** to drive faster toward destination.

AI/ML Example - logistic regression , convex optimization problems

Damped Newton Method : Modified Newton method. $x_{k+1} = x_k - \alpha H^{-1} \nabla f$, where $\alpha < 1$ to avoid overshooting.

Real-Life Example - Driving downhill with **controlled braking**.

AI/ML Example - Used when Newton method becomes unstable.

Quasi-Newton Methods : Newton requires computing Hessian , But Hessian is expensive . So we **approximate Hessian**.

Method - Instead of computing H, we update an approximation B_k .

1) **Davidon-Fletcher-Powell (DFP)** : One of the first quasi-Newton methods

Formula :

$$B_{k+1} = B_k + \frac{ss^T}{s^T y} - \frac{B_k y y^T B_k}{y^T B_k y}$$

Where $s = x_{k+1} - x_k$, $y = \nabla f_{k+1} - \nabla f_k$,

Real-Life Example : Improving a **map of terrain while walking** , You refine curvature estimates as you move.

AI/ML Example : Used in optimization libraries.

1) **BFGS Method**: Improved version of DFP, most widely used quasi-Newton method

Why BFGS is used more ? Reason : Stable , Fast convergence ,

Good for medium-scale problems

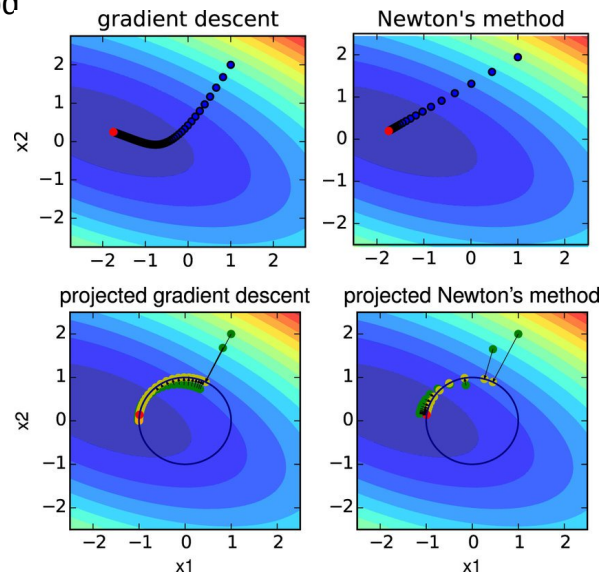
Real-Life Example : GPS navigation improving **route estimation** after each step.

AI/ML Example: scikit-learn optimization , logistic regression ,

support vector machines

Variant : **L-BFGS** used for large datasets

(BFGS : Broyden-Fletcher-Goldfarb-Shanno)



Mathematical approach (Explaining Difference in all the Above Methods) :

Let : $f(x) = x^2$, Goal: $\min f(x)$, Derivative: $\nabla f(x) = 2x$, Start from: $x_0 = 4$, Learning rate: $\alpha = 0.1$

SGD (Stochastic Gradient Descent)

Rule : $x_{k+1} = x_k - \alpha (2x_k)$

Iteration 1 :- Gradient $g = 2(4) = 8$, Update $x_1 = 4 - 0.1 (8)$, $x_1 = 3.2$

Iteration 2 :- $g = 2 (3.2) = 6.4$, $x_2 = 3.2 - 0.1 (6.4)$, $x_2 = 2.56$

Conclusion - SGD slowly approaches minimum

Mini-Batch Gradient Descent

Lets take dataset has 4 samples :

Mini-batch size = 2

Average gradient: $g = (2+4) / 2 = 3$

Now Update : $x_1 = 4 - 0.1 (3)$, $x_1 = 3.7$

Conclusion - Mini-batch gives **more stable update than SGD**.

x	gradient
1	2
2	4
3	6
4	8

Momentum

Velocity formula : $v_t = \beta v_{t-1} + \alpha g$, Let $\beta = 0.9$

Iteration 1 : $g = 8$, $v_1 = 0.9 (0) + 0.1 (8)$, $v_1 = 0.8$

Now Update : $x_1 = 4 - 0.8$, $x_1 = 3.2$

Iteration 2 : $g = 6.4$, $v_2 = 0.9 (0.8) + 0.1 (6.4)$, $v_2 = 0.72 + 0.64$, $v_2 = 1.36$

Now Update : $x_2 = 3.2 - 1.36$, $x_2 = 1.84$

Conclusion - Momentum moves **faster than SGD**

AdaGrad

Learning rate becomes -

$$\alpha_t = \frac{\alpha}{\sqrt{G_t}}$$

Iteration 1 : $g = 8$, $G = 8^2 = 64$

Effective learning rate : $\alpha = 0.1 / 8$, $\alpha = 0.0125$

Now Update : $x_1 = 4 - 0.0125 (8)$, $x_1 = 3.9$

Conclusion - Learning rate decreases

RMSProp

Uses moving average of squared gradient.

Let assume : $\beta = 0.9$

Iteration 1 : $g = 8$, $E[g^2] = 0.9(0) + 0.1(64)$, $E[g^2] = 6.4$

Now Update : $x_1 = 4 - [0.1(8) / \text{Sqrt } 6.4]$, $x_1 \approx 4 - 0.316$, $x_1 \approx 3.684$

Conclusion - RMSProp keeps learning rate **stable**.

Adam Optimizer (Uses two moments)

First moment: $m = \beta_1 m + (1 - \beta_1)g$, Second moment : $v = \beta_2 v + (1 - \beta_2)g^2$

Let take : $\beta_1 = 0.9$, $\beta_2 = 0.999$

Iteration 1 : $g = 8$, $m = 0.8$, $v = 0.064$

Now Update : $x_1 = 4 - [0.1(0.8) / \text{sqrt } 0.064]$, $x_1 \approx 3.0$

Conclusion - Adam converges **very fast**.

Pure Newton Method

Formula : $x_{k+1} = x_k - [f'(x) / f''(x)]$

We take : $f(x) = x^2$, $f'(x) = 2x$, $f''(x) = 2$

Iteration 1 : $x_1 = 4 - [8 / 2]$, $x_1 = 0$

Conclusion - Newton reaches **minimum in one step**.

Damped Newton Method

adding step size: $x_{k+1} = x_k - \alpha (f' / f'')$

Let $\alpha = 0.5$,

Iteration 1 : $x_1 = 4 - 0.5 (4)$, $x_1 = 2$

Iteration 2 : $x_2 = 2 - 0.5 (2)$, $x_2 = 1$

Conclusion - More stable than pure Newton.

Quasi Newton Method:

Newton requires Hessian: H

Since computing Hessian is expensive.

Quasi-Newton **approximates Hessian**.

DFP Method (Numerical)

Let $x_0 = 4$, Gradient : $g_0 = 8$, Shifting to : $x_1 = 3$, New gradient : $g_1 = 6$,

We Compute $s = x_1 - x_0 = -1$, $y = g_1 - g_0 = -2$

DFP updates approximate Hessian using -

$$B_{k+1} = B_k + \frac{ss^T}{s^T y} - \frac{B_k y y^T B_k}{y^T B_k y}$$

Conclusion - It improves curvature estimate each step

BFGS Method

More stable update:

$$B_{k+1} = B_k - \frac{B_k s s^T B_k}{s^T B_k s} + \frac{y y^T}{y^T s}$$

So if : $s = -1$, $y = -2$

Conclusion - New Hessian estimate improves search direction ,

BFGS converges much faster than gradient descent.

Real AI/ML Example - Loss $L(\theta)$, Optimization : $\theta_{k+1} = \theta_k - \alpha \nabla L$

AI model	Optimizer
CNN	Momentum / Adam
RNN	RMSProp
Transformers	Adam
Logistic Regression	BFGS

Questions

1) $f(x) = x^2 + 2x$, Ans : $x_1 = 1.4$, $x_2 = 0.92$

1) $f(x) = x^2$, Ans : $x_1 = 2.4$, $x_2 = 1.44$

1) $f(x) = x^2 + 4x + 4$, Ans : $x_1 = -2$

1) Apply Newton Method for : $x^2 + y^2$, and Find its Solutions.

Week 6

- Conjugate Direction Method
- Exact Solution for the Quadratic Equation
- Fletcher Reeves Conjugate Gradient Method

Conjugate Direction Method

Approach - Instead of moving only along the **negative gradient (steepest descent)**, we move along **special directions called conjugate directions**.

Mathematical Definition : Directions d_i and d_j are **conjugate w.r.t matrix A** if: $d_i^T A d_j = 0$, ($i \neq j$)

Why Needed?

Reason : Gradient descent is Zig-zags , Slow convergence while Conjugate directions remove zig-zag ,reach minimum faster.

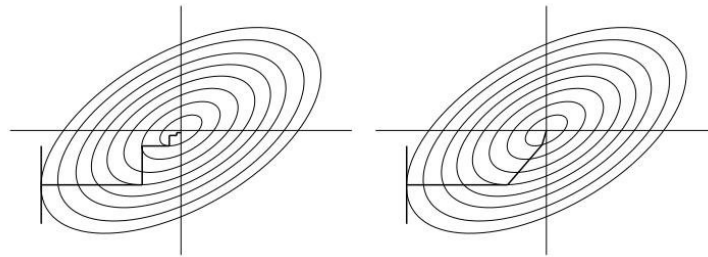
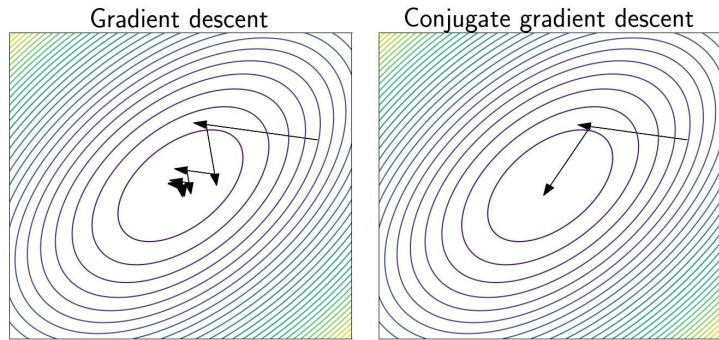


Figure 6.14: Steepest descent vs. conjugate gradient.

Real-Life Example: Imagine walking in valley [Gradient descent → zig-zag walking, Conjugate direction → smart diagonal shortcuts]

AI/ML Connection : Used in - Large linear systems , Quadratic loss problems , Training ML models without computing Hessian.

Kernel methods (SVM).

Algorithm (Conjugate Direction Method):

Start with x_0 , For $k = 0, 1, 2, \dots$

- 1) Choose direction d_k such that: $d_k^T A d_k = 0$
- 2) Compute step size (exact line search) : $\alpha_k = [d_k^T (b - A x_k)] / [d_k^T A d_k]$
- 3) Update : $x_{k+1} = x_k + \alpha_k d_k$

Result - The method converges in **at most n steps** (for n - dimensional problem)

Each direction removes error in one dimension

Exact Solution for Quadratic Function

- 1) Function : $f(x) = \frac{1}{2} x^T A x - b^T x$
- 2) Gradient : $\nabla f(x) = A x - b$
- 3) Optimality Condition : Set gradient to zero - $A x^* - b = 0$, $A x^* = b$
- 4) Exact Solution : $x^* = A^{-1} b$

Conditions: A must be invertible

Interpretation : We solve a **system of linear equations** instead of iterative optimization.

Real-Life Example : Electrical Circuit Analysis [Ohm's law system: $A x = b$, where $x \rightarrow$ currents , $A \rightarrow$ resistance matrix]

AI/ML Connection : Linear Regression [$w = (X^T X)^{-1} X^T y$]

This is exactly the **quadratic optimization solution**.

Fletcher-Reeves Conjugate Gradient Method

A **practical algorithm** to generate conjugate directions **without computing A** explicitly.

Algorithm :

Step 1: Initialization -

$$x_0 \text{ given}, g_0 = \nabla f(x_0), d_0 = -g_0$$

Step 2: Iteration -

(a) Step size : $\alpha_k = (g_k^T g_k) / (d_k^T A d_k)$

(b) Update solution : $x_{k+1} = x_k + \alpha_k d_k$

(c) New gradient : $g_{k+1} = \nabla f(x_{k+1})$

(d) Compute β : $\beta_k = (g_{k+1}^T g_{k+1}) / (g_k^T g_k)$

(e) New direction : $d_{k+1} = -g_{k+1} + \beta_k d_k$

Properties :

- 1) Directions are **automatically conjugate**
- 2) No need to compute Hessian
- 3) Converges in at most n steps (for quadratic case)
- 4) Combines: Gradient (current slope) + New Direction [Smarter than gradient descent]

Real-Life Example :

Navigation System - Gradient descent $\rightarrow$ follow steepest slope

- Fletcher-Reeves $\rightarrow$ remembers past movement and avoids repeating mistakes

AI/ML Connection: Used in Large-scale optimize, Train ML models with huge data, Solving linear systems (Gaussian Process, Kernel methods)

Numericals

1) Conjugate Direction Method (Independent smart directions)

Minimize: $f(x) = \frac{1}{2} x^T A x - b^T x$.

Where $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$, Start with $x_0 = (0, 0)$

Use **two conjugate directions** : $d_0 = (1, 0)$, $d_1 = (0, 1)$

Solution :

Step 1: First Direction

$$r_0 = b - A x_0 = (8, 4)$$

$$\alpha_0 = d_0^T r_0 / d_0^T A d_0$$

$$a_0 = 8/4 = 2$$

Update : $x_1 = x_0 + 2(1, 0) = (2, 0)$

Step 2: Second Direction

$$r_1 = b - A x_1$$

$$A x_1 = (8, 0)$$

$$r_1 = (8, 4) - (8, 0) = (0, 4)$$

$$\alpha_1 = d_1^T r_1 / d_1^T A d_1, \alpha_1 = 4/2 = 2$$

Update : $x_2 = (2, 0) + 2(0, 1) = (2, 2)$

Final Answer : $x^* = (2, 2)$. What we Compute ? Ans: **α and updates**

2) Exact Solution for Quadratic Equation (Solve system directly)

Problem : Solve $\frac{1}{2} x^T A x - b^T x$, where $A = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$

Solution :

Step 1: Optimal condition $Ax = b$

Step 2: Solve system $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$

Step 3: Equations $2 x_1 = 6 \Rightarrow x_1 = 3$

$$4 x_2 = 8 \Rightarrow x_2 = 2$$

Final Answer $x^* = (3, 2)$. What we Compute ? Ans: Solve $Ax = b$

3) Fletcher-Reeves Conjugate Gradient Method (Efficient practical method)

Problem : Minimize $f(x,y) = x^2 + y^2$

Start : $x_0 = (1, 2)$

Solution :

Step 1: Gradient

$$g_0 = (2, 4)$$

$$d_0 = (-2, -4)$$

Step 2: Step size

$$\alpha_0 = (g_0^T g_0) / (d_0^T A d_0)$$

Here: $A = 2 I$

$$g_0^T g_0 = 4 + 16 = 20$$

$$d_0^T A d_0 = 2 (4+16) = 40$$

$$\alpha_0 = 20/40 = 0.5$$

Step 3: Update

$$x_1 = (1, 2) + 0.5 (-2, -4)$$

$$x_1 = (0, 0)$$

Step 4: New Gradient

$$g_1 = (0, 0)$$

Step 5: Compute β

$$\beta_0 = g_1^T g_1 / g_0^T g_0 = 0$$

Step 6: New direction

$$d_1 = -g_1 + 0 \cdot d_0 = 0$$

Final Answer : $x^* = (0, 0)$. What we Compute ? Ans: α, β , direction

Practise Questions :

1) Conjugate Direction Method

Minimize: $f(x) = \frac{1}{2} x^T A x - b^T x$, where $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$, Given : $x_0 = (0,0)$, $d_0 = (1,0)$
Find : α_0 , x_1

1) Using same problem above, take $d_1 = (0,1)$, Find : α_1 , x_2

2) Check whether the directions - $d_0 = (1,0)$, $d_1 = (0,1)$, are conjugate w.r.t $A = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$ Verify : $d_0^T A d_1$?

4) Exact Solution of Quadratic Function

Solve : $Ax = b$. Find x^* ?
 $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$

5) Find the minimum of: $f(x,y) = x^2 + 2y^2 - 4x - 8y$

Find gradient , Set $\nabla f = 0$

Week-7

1. Constrained Optimisation
2. Concept of Feasible and Descent Direction
3. Geometrical Optimality
4. Fritz John Problem

Constrained Optimization

General Problem :- $\min f(\mathbf{x})$, subject to: $\mathbf{g}_i(\mathbf{x}) \leq 0$ ($i = 1, \dots, m$) , $\mathbf{h}_j(\mathbf{x}) = 0$ ($j = 1, \dots, p$)

Types : **Inequality constraints** $\rightarrow g(\mathbf{x}) \leq 0$

Equality constraints $\rightarrow h(\mathbf{x}) = 0$

Feasible Set : $\Omega = \{ \mathbf{x} \mid \mathbf{g}_i(\mathbf{x}) \leq 0, \mathbf{h}_j(\mathbf{x}) = 0 \}$ [**All points satisfying constraints.**]

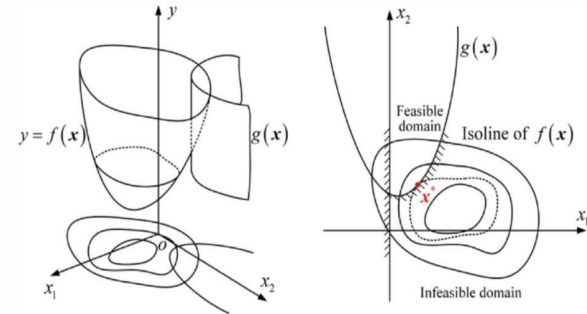
Idea - We want to **optimize**, but only within allowed region.

Real-Life Example : Budget-Constrained Shopping

Minimize cost : **min Cost ($\mathbf{x}$)** , Subject to : Budget ≤ 5000

AI/ML Connection : Used in - Support Vector Machines (SVM) , Regularization constraints , Resource-constrained ML models.

What is Constrained Optimization



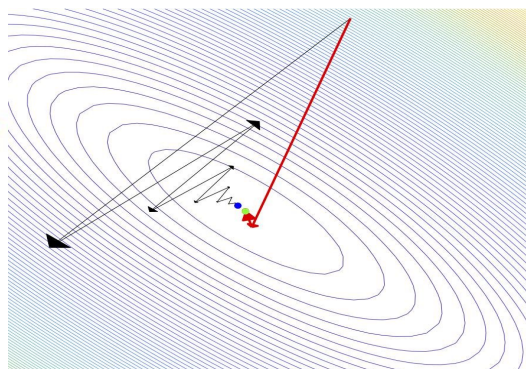
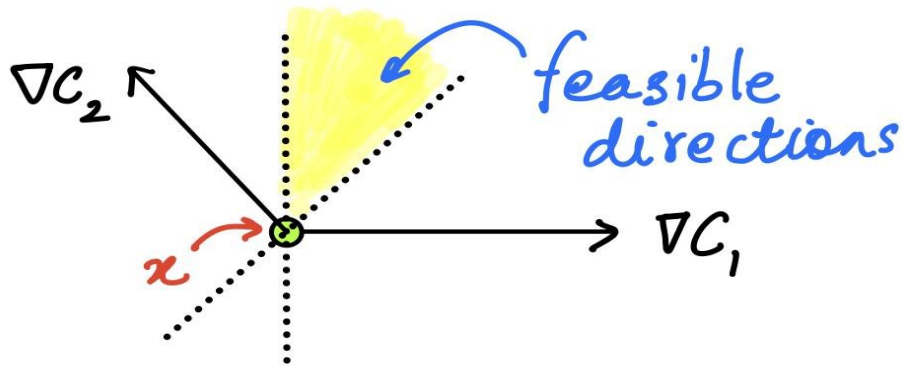
Feasible Direction & Descent Direction

Feasible Direction : A direction $\mathbf{d}$ is feasible at point $\mathbf{x}$ if moving slightly in that direction keeps us inside constraints.

For inequality constraint : $\nabla \mathbf{g}_i(\mathbf{x})^T \mathbf{d} \leq 0$

Descent Direction : A direction $\mathbf{d}$ is descent direction if : $\nabla \mathbf{f}(\mathbf{x})^T \mathbf{d} < 0$

Mathematical Meaning : Function value decreases along that direction.



Idea : We want a direction that is - Feasible , Descent

Real-Life Example - Driving inside a road: Road $\rightarrow$ feasible region ,

Moving downhill $\rightarrow$ descent direction

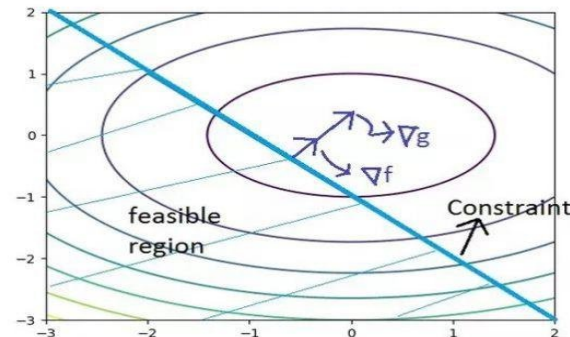
AI/ML Connection : Used in - Projected Gradient Descent , Constrained neural networks , Adversarial robustness training

Geometrical Optimality Condition

Idea : At optimal point - No feasible direction can reduce the function.

Mathematical Condition : If x^* is optimal - $\nabla f(x^*)^T d \geq 0 \forall d \in \text{feasible directions}$

Interpretation : We cannot go **downhill** without leaving feasible region.
We are at "best possible point within constraints"

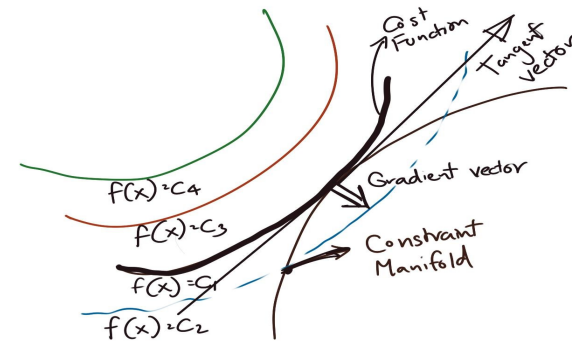


Idea : At optimum - Gradient is **blocked by constraint boundary**.

Real-Life Example : Water flowing in a container - It settles at lowest point ,

But constrained by container walls.

AI/ML Connection : Used in - SVM margin maximization , Constrained optimization layers
Reinforcement learning constraints



Fritz John Conditions

General optimality conditions for constrained optimization. More general than **KKT conditions**. [Karush-Kuhn-Tucker (KKT) conditions are first-order necessary conditions for a solution to be optimal in nonlinear programming, extending Lagrange multipliers to inequality constraints.]

Problem : $\min f(x)$, subject to : $g_i(x) \leq 0$

Fritz John Conditions : There exist multiplier - $\lambda_0, \lambda_1, \dots, \lambda_m$, such that :

1) Stationarity :
$$\lambda_0 \nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) = 0$$

2) Non-negativity : $\lambda_i \geq 0$

3) Complementary Slackness : $\lambda_i g_i(x^*) = 0$

4) Not all multipliers zero : $(\lambda_0, \lambda_1, \dots, \lambda_m) \neq 0$

Special Case (KKT) : If $\lambda_0 = 1$, Then Fritz John $\rightarrow$ **KKT conditions**

At optimum : Objective gradient + constraint gradients balance each other.

Real-Life Example : Balancing forces - Objective → pulling in one direction

Constraints → opposing forces. (System stabilizes at equilibrium)

AI/ML Connection : Used in - SVM optimization , Lagrangian dual problems , Constrained deep learning

Concepts	Meaning	Simple Idea	Key Formula
Constrained Optimization	Optimize with Restrictions	Boundary	$g(x) \leq 0$
Feasible Direction	Stay inside Region	Allowed Movement	$\nabla g^T d \leq 0$
Descent Direction	Reduce Function	Downhill Movement	$\nabla f^T d < 0$
Geometric Optimality	No better Feasible Move	Best Point within Boundary	$\nabla f^T d \geq 0$
Fritz John	General Optimality	Balance of Forces	$\Lambda_0 \nabla f + \sum \lambda_i \nabla g_i = 0$

Numerical Sample Question

1) Constrained Optimization (Basic Problem)

Problem : $f(x,y) = x^2 + y^2$, Subject to: $x+y-1 = 0$

Solution (Using Lagrange Method)

Step 1: Lagrangian : $L = x^2 + y^2 + \lambda (x+y-1)$

Step 2: Partial derivatives : $\partial L / \partial x = 2x + \lambda = 0$, $\partial L / \partial y = 2y + \lambda = 0$, $x+y-1 = 0$

Step 3: Solve : From first two : $2x = 2y \Rightarrow x = y$, Substitute : $x+x = 1 \Rightarrow x = \frac{1}{2}$, $y = \frac{1}{2}$

Final Answer :- $(x^*, y^*) = (\frac{1}{2}, \frac{1}{2})$

1) Feasible Direction

Problem : Given constraint : $g(x,y) = x+y-1 \leq 0$. At point $(0,1)$, Check whether direction : $d=(1,-1)$ is feasible.

Solution

Step 1: Gradient : $\nabla g = (1, 1)$

Step 2: Feasibility condition : $\nabla g^T d \leq 0 \Rightarrow (1,1) \cdot (1,-1) = 1-1 = 0$

Since: $0 \leq 0$

Direction is feasible.

3) Descent Direction

Problem : Given : $f(x,y) = x^2 + y^2$. At point: $(1,1)$, Check whether : $d = (-1,0)$ is a descent direction.

Solution

Step 1: Gradient : $\nabla f = (2x, 2y) = (2, 2)$

Step 2: Check condition : $\nabla f^T d < 0 \Rightarrow (2, 2) \cdot (-1, 0) = -2$

Since: $-2 < 0$

Direction is **descent direction**.

4) Geometrical Optimality

Problem :- Minimize : $f(x,y) = x^2 + y^2$. Subject to: $x+y \leq 1$, Check whether point: $(0.5, 0.5)$ is optimal.

Solution

Step 1: Gradient : $\nabla f = (2x, 2y) = (1, 1)$

Step 2: Constraint gradient : $\nabla g = (1, 1)$

Step 3: Check condition : At boundary : $\nabla f = \lambda \nabla g \Rightarrow (1, 1) = \lambda(1, 1) \Rightarrow \lambda = 1$

Condition satisfied $\rightarrow$ optimal.

5) Fritz John Problem

Problem : Minimize : $f(x,y) = x^2 + y^2$. Subject to: $g(x,y) = x+y-1 \leq 0$, Find optimal solution using **Fritz John conditions**.

Solution

Step 1: Gradients : $\nabla f = (2x, 2y)$, $\nabla g = (1, 1)$

Step 2: Fritz John condition : $\lambda_0 \nabla f + \lambda_1 \nabla g = 0 \Rightarrow \lambda_0 (2x, 2y) + \lambda_1 (1, 1) = 0$

Step 3: Solve : $2 \lambda_0 x + \lambda_1 = 0 \Rightarrow 2 \lambda_0 y + \lambda_1 = 0$

Subtract: $x = y$

Step 4: Constraint : $x+y = 1$, $2x = 1 \Rightarrow x = \frac{1}{2}$, $y = \frac{1}{2}$

Step 5: Multipliers : $\lambda_1 = -2 \lambda_0$, $x = -\lambda_0$

Final Answer : $(x^*, y^*) = (\frac{1}{2}, \frac{1}{2})$

Fritz John (Concept Check Problem)

Problem : Check if point satisfies Fritz John : $f(x) = x^2$, $g(x) = x-2 \leq 0$, At $x^* = 2$

Solution

Step 1: Gradients : $\nabla f = 2x = 4$, $\nabla g = 1$

Step 2: Condition : $\lambda_0(4) + \lambda_1(1) = 0$, Choose:- $\lambda_0 = 1$, $\lambda_1 = -4$, Check feasibility: $g(2) = 0$,

Fritz John satisfied

Questions

1)Constrained Optimization

Q1 - Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x+y = 2$. Find optimal x , y .

Q2 - Minimize: $f(x,y) = x^2 + y^2$, Subject to : $x = 1$. Find optimal y .

2)Feasible Direction

Q3 - Given constraint: $g(x,y) = x+y-2 \leq 0$. At point : $(1,1)$, Check if direction: $d=(1,0)$ is feasible.

Q4 - For same constraint above, check whether: $d = (-1,-1)$, is feasible.

Q5 - $g(x,y)=x^2 + y^2 - 4 \leq 0$. At point: $(1,1)$, Check whether: $d = (1,1)$, is feasible

Do the Descent Direction & Geometrical Optimality of the above Numericals

3) Fritz John Conditions

Q7 - Given : $f(x,y) = x^2 + y^2$, $g(x,y) = x+y-2 \leq 0$. Write Fritz John condition equation.

Q8 - Given gradients: $\nabla f = (2,2)$, $\nabla g = (1,1)$. Write multiplier equation.

Week 8

- KKT Conditions
- SVM (Support Vector Machine)
- Fritz John Conditions (Unconstraint Optimization)

KKT (Karush–Kuhn–Tucker) Conditions

Problem (General Constrained Optimization) - $\min f(\mathbf{x})$, subject to : $g_i(\mathbf{x}) \leq 0$ ($i = 1, \dots, m$) , $h_j(\mathbf{x}) = 0$ ($j = 1, \dots, p$)

Lagrangian :
$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \mu_j h_j(\mathbf{x})$$

$\lambda_i \rightarrow$ multipliers for inequalities

$\mu_j \rightarrow$ multipliers for equalities

KKT Conditions : At optimum $\mathbf{x}^*$, there exist λ_i, μ_j such that :

1) Stationarity :
$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \mu_j \nabla h_j(\mathbf{x}^*) = 0$$

2) Primal Feasibility : $g_i(\mathbf{x}^*) \leq 0$, $h_j(\mathbf{x}^*) = 0$

3) Dual Feasibility : $\lambda_i \geq 0$

4) Complementary Slackness : $\lambda_i g_i(\mathbf{x}^*) = 0$

Geometric Meaning : a) The **objective gradient** is balanced by **active constraints' gradients**.

b) Only **active constraints** (where $g_i(\mathbf{x}^*) = 0$) influence the solution.

Real-Life Example : Diet optimization - Minimize cost while meeting nutrition constraints.

At optimum - Some constraints are active (e.g., minimum protein met exactly)

Others inactive (e.g., vitamin limit not tight)

AI/ML Connection : Core to **SVM training** , Used in **regularization (L1/L2 constraints)** , Appears in **resource-constrained ML (edge devices)**

Support Vector Machine (SVM)

Aim : Find a **maximum margin hyperplane** that separates two classes.

a) Hard-Margin SVM (Linearly Separable)

Given data (x_i, y_i) , $y_i \in \{-1, +1\}$: $\min_{w,b} \frac{1}{2} \|w\|^2$, subject to : $y_i (w^T x_i + b) \geq 1$

a) Lagrangian (Using KKT) : $\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i [y_i (w^T x_i + b) - 1]$

c) KKT Conditions in SVM

i) Stationarity :

$$w = \sum_{i=1}^n \alpha_i y_i x_i$$

ii) Complementary Slackness : $\alpha_i [y_i (w^T x_i + b) - 1] = 0$

iii) $\alpha_i \geq 0$

Key Idea : Only points with $\alpha_i > 0$ matter $\rightarrow$ **Support Vectors**

Real-Life Example : Email Spam Detection

Emails $\rightarrow$ data points

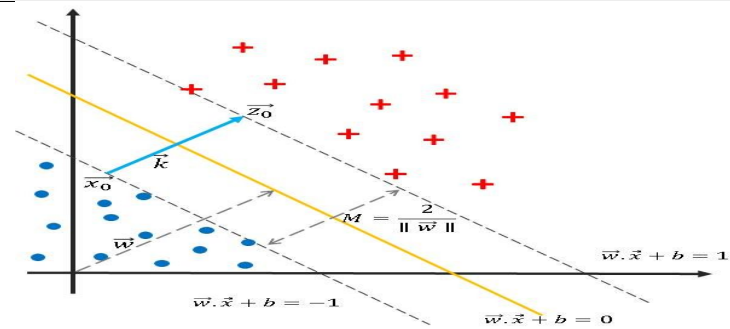
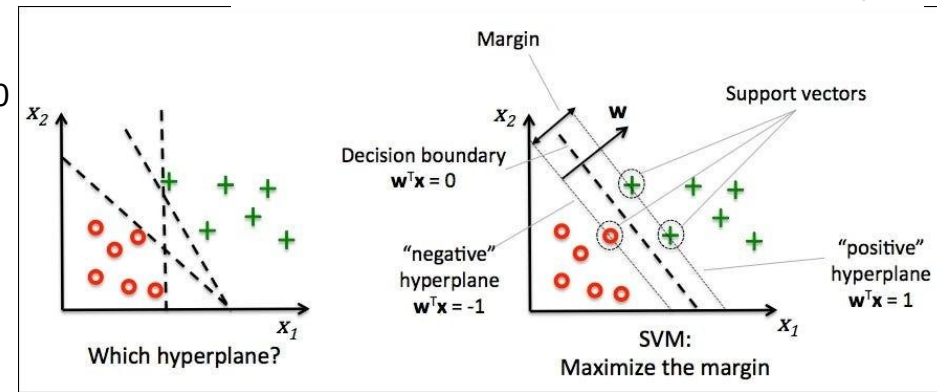
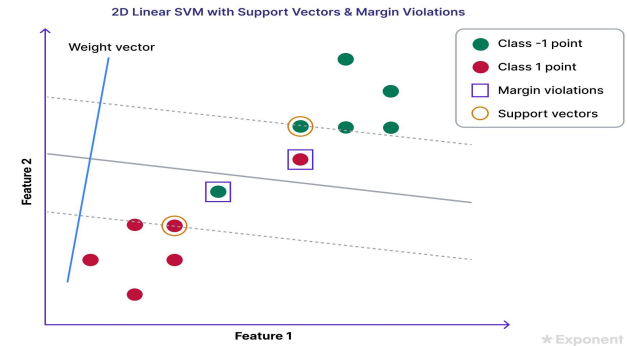
Boundary $\rightarrow$ separates spam vs non-spam

Support vectors $\rightarrow$ critical borderline emails

AI/ML Connection : SVM is used in

Text classification , Face detection ,

Bioinformatics , Kernel methods (non-linear SVM)



Fritz John Conditions

General Form (Constrained) : $\min f(x)$, $g_i(x) \leq 0$

Fritz John Conditions : There exist multipliers: $\lambda_0, \lambda_1, \dots, \lambda_m$, such that :

1) Stationarity :
$$\lambda_0 \nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) = 0$$

2) Dual Feasibility: $\lambda_i \geq 0$

3) Complementary Slackness : $\lambda_i g_i(x^*) = 0$

4) Non-triviality : $(\lambda_0, \lambda_1, \dots, \lambda_m) \neq 0$

Relation to KKT : If $\lambda_0 = 1 \rightarrow$ becomes **KKT conditions** , Fritz John is more **general** (does not require constraint qualification)

Special Case: **Unconstrained Optimization**

If no constraints: $\min f(x)$, Then: $\nabla f(x^*) = 0$, This is **Fritz John reduced form**.

Real-Life Example : Force balance system

[Objective $\rightarrow$ one force , Constraints $\rightarrow$ opposing forces , At equilibrium $\rightarrow$ all forces balance]

AI/ML Connection : Used in **Lagrangian dual problems** , Basis of **SVM optimization theory** ,

Appears in **deep learning constrained training**

TABLE

Concept	Meaning	Equation	Use
KKT	Balance between objective & constraints	$\nabla f + \sum \lambda \nabla g = 0$	Standard constrained optimization
SVM	Best separating boundary	$\min \frac{1}{2} \ w\ ^2$	Classification
Fritz John	General condition for optimality	$\Lambda_0 \nabla f + \sum \lambda_i \nabla g_i = 0$	General optimality
Unconstrained	Flat slope at optimum	$\nabla f = 0$	Simple optimization

NUMERICAL SOLUTION

1) KKT Conditions (Equality Constraint)

Problem - Minimize: $f(x,y) = x^2 + y^2$, Subject to : $x + y = 1$

Solution :

Step 1: Lagrangian : $L = x^2 + y^2 + \lambda(x + y - 1)$

Step 2: Stationarity : $\partial L / \partial x = 2x + \lambda = 0$, $\partial L / \partial y = 2y + \lambda = 0$

Step 3: Solve : $2x = 2y \Rightarrow x = y$, $x + y = 1 \Rightarrow 2x = 1$, $x = 1/2$, $y = 1/2$

Final Answer: $(x^*, y^*) = (1/2, 1/2)$

2) KKT Conditions (Inequality Constraint)

Problem - Minimize : $f(x) = x^2$, Subject to : $g(x) = x - 1 \geq 0$

Solution: Rewrite constraint: $g(x) = 1 - x \leq 0$

Step 1: Lagrangian : $L = x^2 + \lambda(1 - x)$

Step 2: Stationarity : $dL/dx = 2x - \lambda = 0$, $\lambda = 2x$

Step 3: Complementary slackness : $\lambda(1 - x) = 0$

Case 1: $x < 1 \Rightarrow$ Not feasible $\rightarrow$ reject

Case 2: $x = 1 \Rightarrow \lambda = 2$

Final Answer : $x^* = 1$

3) SVM (Hard Margin)

Problem : Given points - Class +1: (2,2),(4,4)

Class -1: (2,0),(4,0) , Find separating hyperplane.

Solution :

Step 1: Assume form : $w^T x + b = 0$, Let: $w = (w_1, w_2)$

Step 2: Constraints : For +1: $w_1 x + w_2 y + b \geq 1$

For -1: $w_1 x + w_2 y + b \leq -1$

Step 3: Observation : Points differ only in **y-direction** , So choose : $w = (0,1)$

Step 4: Find b , For (2,2) : $2 + b = 1 \Rightarrow b = -1$,

Final Hyperplane : $y - 1 = 0$

Answer: $w = (0,1)$, $b = -1$

4) SVM (Support Vectors Identification)

Problem : Given hyperplane: $y = 1$

Find support vectors from : (2,2), (4,4), (2,0), (4,0)

Solution : Support vectors lie on: $y = 2$ and $y = 0$, Closest points : (2,2), (2,0)

Answer : Support vectors = (2,2),(2,0)

5) Fritz John Conditions

Problem : Minimize: $f(x,y) = x^2 + y^2$, Subject to : $g(x,y) = x + y - 2 \leq 0$

Solution :

Step 1: Gradients : $\nabla f = (2x, 2y)$, $\nabla g = (1,1)$

Step 2: Fritz John equation : $\lambda_0 (2x, 2y) + \lambda_1 (1,1) = 0$

Step 3: Solve : $2\lambda_0 x + \lambda_1 = 0$, $2\lambda_0 y + \lambda_1 = 0 \Rightarrow$ Subtract : $x = y$

Step 4: Constraint : $x + y = 2$, $2x = 2 \Rightarrow x = 1$, $y = 1$

Step 5: Multipliers : $\lambda_1 = -2 \lambda_0$

Final Answer : $(x^*, y^*) = (1,1)$

6) Fritz John (Unconstrained Case)

Problem : Minimize : $f(x) = x^2 - 4x$

Solution :

step 1: Gradient : $df/dx = 2x - 4$

Step 2: Set zero : $2x - 4 = 0$, $x = 2$

Final Answer : $x^* = 2$

Practise Question

A) KKT

- 1) Minimize: $f(x) = x^2$, Subject to: $x \geq 1$, Find optimal x using KKT.
- 2) Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x+y = 2$, Find optimal x,y .
- 3) Write the **Lagrangian** for: $f(x,y) = x^2 + y^2$, Subject to: $x+y-1 \leq 0$

B) SVM

- 1) Given two points: Class +1: (2,2)
Class -1: (2,0) , Suggest a separating line.
- 2) Given hyperplane: $y = 1$, Classify point (2,2) and (2,0)

C) Fritz John Conditions

- 1) Write Fritz John condition for: $\min f(x)$, subject to : $g(x) \leq 0$

D) Unconstrained Optimization (Fritz John Case)

- 1) Find minimum of: $f(x) = x^2 - 4x$
- 2) Find stationary point of: $f(x) = x^2 + 2x + 1$

Week 9

KKT

SVM explained in detailed in the recorded classes

Week 10

- Numerical methods for Constrained Optimisation
- Lagrange Multiplier Method, Penalty Method
- Quadratic Penalty Function and its Variants: Exact Penalty Function

Lagrange Multiplier Method

Equality Constraints (Problem) : $\min \mathbf{f}(\mathbf{x})$ subject to $\mathbf{h}(\mathbf{x}) = 0$

Lagrangian : $L(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda h(\mathbf{x})$

Optimality Conditions : $\nabla_{\mathbf{x}} L = \nabla f(\mathbf{x}) + \lambda \nabla h(\mathbf{x}) = 0$, $h(\mathbf{x}) = 0$

GOAL (or Idea) is : $\nabla f \rightarrow$ direction of improvement , $\nabla h \rightarrow$ constraint boundary , At optimum : $\nabla f = -\lambda \nabla h$, Gradients are **parallel**

Real-Life Example : Maximizing output with fixed resources

i) Maximize profit , ii) Constraint: limited budget , At optimum $\rightarrow$ we can't improve profit without violating budget.

AI/ML Connection : Used in **SVM dual formulation** , **Equality constraints** in optimization , **Physics-informed ML** models.

Penalty Method

Goal : Convert **constrained problem** into **unconstrained problem**.

Problem : $\min \mathbf{f}(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) \leq 0$

Penalty Function states : $F(\mathbf{x}, r) = f(\mathbf{x}) + r \cdot P(\mathbf{x})$

Where : r = penalty parameter , $P(\mathbf{x})$ = penalty for violating constraints

Idea : If constraints are violated $\rightarrow$ heavy penalty , so it Forces solution toward feasible region.

Real-Life Example : Traffic rules - Driving fine (penalty) increases if you break rules (Encourages staying within constraints)

AI/ML Connection : Constrained neural networks , Regularization (L1, L2) , Reinforcement learning with penalties.

Quadratic Penalty Function

For Equality Constraint : $h(x) = 0$, Penalty function : $F(x,r) = f(x) + r [h(x)]^2$

For Inequality Constraint : $g(x) \leq 0$, Penalty function : $F(x,r) = f(x) + r [\max(0, g(x))]^2$

Algorithm Steps : 1) Choose small r 2) Minimize $F(x,r)$

3) Increase $r \rightarrow \infty$ 4) Repeat

Goal : Squared penalty $\rightarrow$ smooth function , But requires large r for accuracy

Real-Life Example : Late submission penalty (Penalty grows **quadratically** with delay.)

AI/ML Connection : Loss functions with penalty terms , Regularized regression : $L = \text{error} + \lambda \|w\|^2$

Exact Penalty Function

Problem : Quadratic penalty needs $r \rightarrow \infty$ (Exact penalty solves this)

Form : For equality constraint : $F(x,r) = f(x) + r |h(x)|$

For inequality : $F(x,r) = f(x) + r \cdot \max(0, g(x))$

Key Property : Solution is exact for **finite r** (No need for $r \rightarrow \infty$)

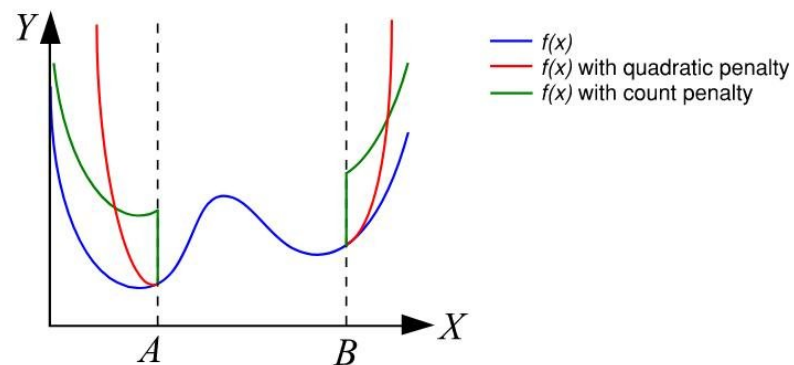
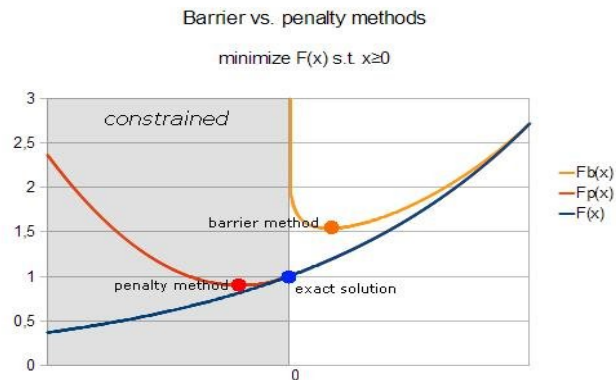
Idea : Linear penalty is **sharper** , Stronger push toward feasibility.

Real-Life Example : Strict penalty system (Immediate penalty if rule violated , No gradual tolerance)

AI/ML Connection : Used in - Sparse optimization (L1 penalty) , Robust loss functions , Constraint enforcement in deep learning

Comparison Table

Method	Formula	Advantage	Limitation	Meaning
Lagrange	$f + \lambda h$	Exact solution	Needs equality constraint	Balance constraint & objective
Penalty	$f + r \cdot P(x)$	Converts to unconstrained	Needs Tuning	Punish constraint violation
Quadratic Penalty	$f + r h^2$	Smooth	Needs large r	Smooth punishment
Exact Penalty	$f + r h$			Sharp strict punishment



Numerical Solution

1) Lagrange Multiplier Method

Problem -: Minimize : $f(x,y) = x^2 + y^2$, Subject to : $x+y = 4$

Solution :

Step 1: Lagrangian : $L = x^2 + y^2 + \lambda (x+y-4)$

Step 2: Partial derivatives : $\partial L/\partial x = 2x+\lambda = 0$, $\partial L/\partial y = 2y+\lambda = 0$, $x+y = 4$

Step 3: Solve : From first two : $2x = 2y \Rightarrow x = y$, Substitute: $x+x = 4 \Rightarrow x = 2$ & $y=2$

Final Answer : $(x^*, y^*) = (2, 2)$

2) Quadratic Penalty Method

Problem -: Minimize : $f(x) = x^2$, Subject to : $x-2 = 0$, Using quadratic penalty with $r = 10$

Solution :

Penalty function : $F(x) = x^2 + 10 (x-2)^2$

Step 1: Expand : $F(x) = x^2 + 10 (x^2 - 4x + 4)$, $F(x) = x^2 + 10 x^2 - 40x + 40$, $F(x) = 11x^2 - 40x + 40$

Step 2: Differentiate : $dF/dx = 22x - 40$

Step 3: Set zero : $22 x = 40$, $x = 40/22 = 1.818$

Final Answer : $x \approx 1.818$

3) Exact Penalty Method

Problem : Minimize : $f(x) = x^2$, Subject to : $x-2 = 0$, Using exact penalty with $r = 5$

Solution :

Penalty function: $F(x) = x^2 + 5 |x-2|$

Case 1: $x \geq 2$

$$F(x) = x^2 + 5(x-2) , F'(x) = 2x + 5$$

$$\text{Set zero : } 2x + 5 = 0 \Rightarrow x = -2.5$$

Not valid ($x \geq 2$)

Case 2: $x < 2$

$$F(x) = x^2 + 5(2-x) , F'(x) = 2x - 5$$

$$\text{Set zero : } 2x - 5 = 0 \Rightarrow x = 2.5$$

Not valid ($x < 2$)

Check boundary : $x = 2$

Final Answer : $x^* = 2$

4) Penalty Method (Inequality Constraint)

Problem : Minimize : $f(x) = x^2$, Subject to: $x-1 \leq 0$, Using penalty : $F(x) = x^2 + r \cdot \max(0, x-1)^2$

Solution :

Case 1: $x \leq 1$

$$F(x) = x^2$$

Minimum at : $x = 0$

Check feasibility : $0 \leq 1 \Rightarrow \text{valid}$

Case 2: $x > 1$

Penalty added $\rightarrow$ higher value

Final Answer : $x^* = 0$

Practise Problem

1) Lagrange Multiplier Method

Q1 Minimize: $f(x,y) = x^2 + y^2 + xy$, Subject to: $x+y = 3$, Find optimal x, y using Lagrange method.

Q2 Minimize: $f(x,y) = x^2 + 4y^2$, Subject to: $2x+y = 5$, Solve completely.

Q3 Minimize: $f(x,y,z) = x^2 + y^2 + z^2$, Subject to: $x + y + z = 6$, Find optimal values.

Q4 (Conceptual) Show that at optimum: $\nabla f = -\lambda \nabla h$, Explain geometrically.

Q5 Minimize: $f(x,y) = x^2 + y^2 - 4x - 6y$, Subject to: $x^2 + y^2 = 5$, Solve using Lagrange multipliers.

Q6 Minimize: $f(x,y) = xy$, Subject to: $x^2 + y^2 = 10$, Find minimum and maximum values.

2) Quadratic Penalty Method

Q7 Minimize: $f(x) = x^2$, Subject to: $x-3 = 0$, Use quadratic penalty with $r = 5$

Q8 Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x + y = 2$, Write penalty function and find approximate solution for $r = 10$

Q9 Minimize: $f(x) = x^2 - 2x$, Subject to: $x-1 = 0$, Solve using penalty method.

Q10 Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x + 2y - 4 = 0$, Solve using quadratic penalty with $r = 50$

Q11 Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x^2 + y^2 - 1 = 0$, Form penalty function and derive equations.

3) Exact Penalty Method

Q12 Minimize: $f(x) = x^2$, Subject to: $x-2 = 0$, Use exact penalty with $r = 4$

Q13 Minimize: $f(x) = x^2 - 4x$, Subject to: $x-1 = 0$, Solve using exact penalty.

Q14 Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x + y - 3 = 0$, Solve using exact penalty (case-wise analysis).

Q15 Minimize: $f(x) = x^2$, Subject to: $x \geq 2$, Use exact penalty: $F(x) = x^2 + r \cdot \max(0, 2-x)$, Solve for $r = 5$

4) Penalty Method (Inequality Constraint)

Q16 Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x+y \leq 2$, Use penalty function: $F(x,y) = x^2 + y^2 + r \cdot \max(0, x+y-2)^2$

Q17 Minimize: $f(x,y) = (x-1)^2 + (y-2)^2$, Subject to: $x+y = 3$

Compare:

- Lagrange solution
- Penalty solution

Week 12's topics

- Barrier function
- Barrier function vs penalty function
- Convergence result

Barrier Function

Equation (Inequality Constraints) : $\min f(x)$ subject to $g_i(x) \leq 0$

Idea : Instead of penalizing violations, we **prevent entering infeasible region** by adding a barrier that becomes infinite near the boundary.

Barrier Function Form : $F(x, \mu) = f(x) + \mu B(x)$

Where
$$B(x) = - \sum_{i=1}^m \ln(-g_i(x))$$

Final Form :
$$F(x, \mu) = f(x) - \mu \sum_{i=1}^m \ln(-g_i(x))$$

Key Property : Defined only for: $g_i(x) < 0$, As $g_i(x) \rightarrow 0$: , $\ln(-g_i(x)) \rightarrow -\infty \Rightarrow F(x) \rightarrow \infty$

Conclusion : i) You are inside feasible region , ii) Boundary acts like a **wall** , iii) You cannot cross it.

Real-Life Example : **Safe driving within road lanes** (Road boundaries $\rightarrow$ constraints , Barrier $\rightarrow$ prevents crossing)

AI/ML Connection : Used in - (Interior-point methods , Constrained optimization in ML , Training models with strict constraints)

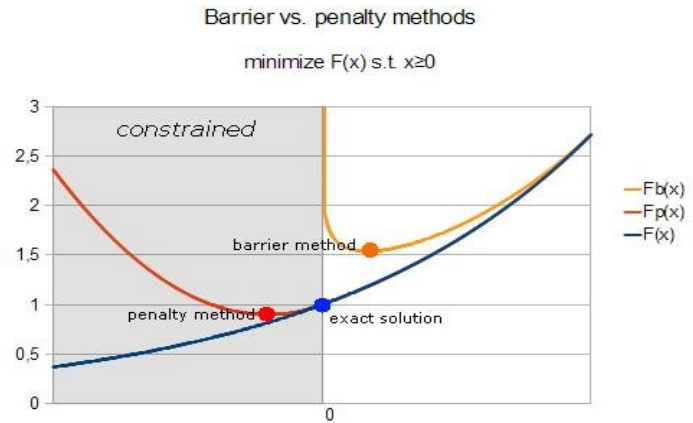
Barrier Function vs Penalty Function

Penalty Function Equation : $F(x, r) = f(x) + r \cdot P(x)$

- a) Allows infeasible points
- b) Penalizes violations

Barrier Function : $F(x, \mu) = f(x) - \mu \sum_{i=1}^m \ln(-g_i(x))$

- a) Does NOT allow infeasible region
- b) Keeps solution strictly inside
 - i) Barrier (inside only)
 - ii) Penalty (outside allowed but punished)



Comparison Table

Feature	Barrier Function	Penalty Function
Feasible region	Inside only	Inside + outside
Constraint violation	Not allowed	Allowed (penalized)
Parameter	$\mu \rightarrow 0$	$r \rightarrow \infty$
Behavior	Pushes away from boundary	Pushes toward feasible region

Conclusion : a) **Barrier** → “Don’t cross the boundary” b) **Penalty** → “You can cross, but pay a cost”

Real-Life Example : Barrier (Security system blocks entry) , Penalty (Fine for breaking rules)

AI/ML Connection : Barrier → strict constraints (robotics, safety ML) , Penalty → soft constraints (regularization)

Convergence Results

Barrier Method Convergence : We solve sequence : $\min F(x, \mu) = f(x) - \mu \sum \ln(-g_i(x))$, with $\mu \rightarrow 0$

Result : $x(\mu) \rightarrow x^*$, where x^* is optimal solution.

Central Path: Sequence of solutions forms: $\{ x(\mu) \}$, called **central path**.

Penalty Method Convergence : $F(x,r) = f(x) + r \cdot P(x)$, with: $r \rightarrow \infty$

Result : $x(r) \rightarrow x^*$

Penalty : Ill-conditioned for large r (push from outside) ; Barrier : Requires feasible start (guide from inside)

Real-Life Example: Barrier - Walking inside room toward door (Stay inside , Gradually approach boundary)

Penalty - WE start outside (Pay penalty , Move toward allowed region)

AI/ML Connection : Used in [Interior point methods (large-scale optimization), Training constrained models ,Reinforcement learning constraints]

Concept	Formula	Key Idea	Meaning
Barrier	$f - \mu \sum \ln (-g)$	Stay inside	Invisible wall
Penalty	$f + r P(x)$	Penalize violation	Fine for Violation
Convergence	$\mu \rightarrow 0$, $r \rightarrow \infty$	Reach optimum	Gradual Movement to Optimum

Numerical Method

Barrier Function (Log Barrier)

Problem : Minimize: $f(x) = x^2$, Subject to: $x - 1 \leq 0$, Using barrier function with $\mu = 1$

Solution :

Step 1: Constraint form : $g(x) = x - 1 \leq 0$, $-g(x) = 1 - x > 0$

Step 2: Barrier function : $F(x) = x^2 - \ln(1-x)$

Step 3: Differentiate : $dF/dx = 2x + 1/(1-x)$

Step 4: Set to zero : $2x + 1/(1-x) = 0$

Multiply: $2x \cdot (1-x) + 1 = 0$, $2x - 2x^2 + 1 = 0$, $2x^2 - 2x - 1 = 0$

Step 5: Solve : $x = (2 \pm \sqrt{4+8}) / 4$, $x = (2 \pm 3.464) / 4$

Feasible solution: $x \approx -0.366$

Final Answer: $x \approx -0.366$

Barrier Function (Different μ)

Problem: Same problem with: $\mu=0.1$

Solution :

Barrier function: $F(x) = x^2 - 0.1 \ln(1-x)$

Derivative: $2x + 0.1 / (1-x) = 0$

Multiply: $2x \cdot (1-x) + 0.1 = 0$, $2x - 2x^2 + 0.1 = 0$, $2x^2 - 2x - 0.1 = 0$

Solve: $x = (2 \pm \sqrt{4 + 0.8}) / 4$, $x = (2 \pm \sqrt{4.8}) / 4$, $x \approx -0.049$

Final Answer : $x \approx -0.049$ (Closer to true solution $x = 0$)

Penalty Method

Problem : Minimize: $f(x) = x^2$, Subject to : $x - 1 \leq 0$, Using penalty: $F(x) = x^2 + r \cdot \max(0, x-1)^2$

Solution:

Case 1: $x \leq 1$

$$F(x) = x^2$$

Minimum : $x = 0$

Case 2: $x > 1$

Penalty increases value

Final Answer: $x^* = 0$

Barrier vs Penalty (Comparison Problem)

Problem: Minimize: $f(x) = (x-2)^2$, Subject to : $x \leq 1$

Solution

Barrier Method: $F(x) = (x-2)^2 - \mu \cdot \ln(1-x)$

Solution stays strictly: $x < 1$

Penalty Method: $F(x) = (x-2)^2 + r \cdot \max(0, x-1)^2$

Solution may approach boundary from outside.

Conclusion : (Barrier $\rightarrow$ inside approach , Penalty $\rightarrow$ outside correction)

Convergence (Barrier Method)

Problem: Given sequence:: $\mu=1,0.1,0.01$, Solutions: $x_1 = -0.36$, $x_2 = -0.049$, $x_3 \rightarrow 0$

Solution: As: $\mu \rightarrow 0$, $x(\mu) \rightarrow 0$

Final Answer: Converges to true minimum : $x^* = 0$

Convergence (Penalty Method)

Problem: Given: $r = 1,10,100$, Solutions: $x_1 = 1.5$, $x_2 = 1.1$, $x_3 \rightarrow 1$

Solution: As: $r \rightarrow \infty$, $x(r) \rightarrow 1$

Final Answer: Converges to boundary solution.

Practise Problem

Barrier Function

- 1) Minimize: $f(x) = x^2$, Subject to: $x-2 \leq 0$, Form the **log barrier function** and find the stationary point for: $\mu=1$
- 2) Minimize: $f(x) = (x-1)^2$, Subject to: $x \leq 0$, Use barrier function and solve for: $\mu = 0.5$
- 3) Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x + y - 1 \leq 0$, Form barrier function and write first-order conditions.
- 4) Show that barrier function is defined only when: $g(x) < 0$, Explain mathematically.

Barrier Function (Hard)

- 5) Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x + y - 2 \leq 0$, Use barrier function: $F(x,y) = x^2 + y^2 - \mu \cdot \ln(2 - x - y)$

Find stationary equations and solve for $\mu = 0.1$

- 6) Minimize: $f(x) = x^2 - 2x$, Subject to: $x \leq 1$, Solve using barrier function for: $\mu = 1, 0.1, 0.01$, Analyze how solution changes.

Penalty vs Barrier (Medium)

- 7) Minimize: $f(x) = (x-3)^2$, Subject to: $x \leq 2$, Write: i) Penalty function ii) Barrier function. Compare both formulations.
- 8) Explain difference between barrier and penalty methods for: $f(x) = x^2$, Subject to: $x \leq 1$, Show mathematical expressions.
- 9) True or False (justify): Barrier method can start from infeasible point.
- 10) Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x + y \leq 2$, Compare: i) Penalty method solution, ii) Barrier method solution

Explain behavior graphically.

Convergence

12) Given barrier sequence: , $\mu = 1, 0.1, 0.01$, Explain how solution approaches optimum.

13) Given penalty sequence: , $r = 1, 10, 100$, Explain convergence behavior.

14) Fill in the blank: , Barrier method converges when: , $\mu \rightarrow \underline{\hspace{1cm}}$

Convergence (Hard)

15) Prove that for barrier function: $F(x, \mu) = f(x) - \mu \cdot \sum \ln(-g(x))$, solution converges to optimum as: $\mu \rightarrow 0$

16) Show that penalty method may become ill-conditioned as: $r \rightarrow \infty$

17) For function: $f(x) = x^2$, Subject to: $x \leq 1$

Compare convergence paths for:

- Barrier method
- Penalty method

Concept + Theory (Important)

18) Explain **central path** in barrier method.

19) Explain why barrier method requires **feasible starting point**.

20) Explain geometrically: Barrier vs Penalty behavior

Week 13

Compactness, characterisation of solution

Duality

Lagrange Dual

Weak duality and Strong Duality(Slater's Conditions)

Complementary Slackness

Compactness & Characterisation of Solution

Definition A set $S \subset \mathbb{R}^n$ is **compact** if: i) It is **closed** (contains boundary points) , ii) It is **bounded** (finite size)

Importance in Optimization : If : $f(x)$ is continuous , S is compact , Then: $\exists x^* \in S$ such that $f(x^*) = \min_{x \in S} f(x)$

Guarantees existence of optimal solution

Conclusion : a) Compact set = “finite, closed region” , b) Optimization inside it always has a solution

Characterisation of Solution : A point x^* is optimal if:

Unconstrained: $\nabla f(x^*) = 0$

Constrained (KKT): $\nabla f(x^*) + \sum \lambda_i \nabla g_i(x^*) = 0$

Geometric condition: $\nabla f(x^*)^T d \geq 0$,for all feasible directions d

Meaning: i) No direction improves solution, ii) Gradient is “blocked” by constraints

Real-Life Example: Budget shopping issue(Budget limits $\rightarrow$ bounded, All options considered $\rightarrow$ closed ,Best purchase always exists.)

AI/ML Connection: Parameter constraints in ML , Ensures loss minimization is achievable , Used in **regularized optimization**.

Characterisation of Solution

Point x^* is optimal if: **Unconstrained:** $\nabla f(x^*) = 0$;

Constrained (KKT): $\nabla f(x^*) + \sum \lambda_i \nabla g_i(x^*) = 0$;

Geometric condition:

$\nabla f(x^*)^T d \geq 0$, (for all feasible directions d)

Meaning : i) No direction improves solution ,
ii) Gradient is “blocked” by constraints

Duality

Aim : Every optimization problem (**primal**) has a corresponding **dual problem**.

Primal Problem: $\min f(x)$, subject to $g_i(x) \leq 0$

Dual Problem: $\max \theta(\lambda)$, where: $\theta(\lambda) = \inf_x L(x, \lambda)$

Key Insight: i) Primal $\rightarrow$ original problem , ii) Dual $\rightarrow$ transformed problem , iii) Solving dual can be easier

Real-Life Example : Company production problem (Primal $\rightarrow$ minimize cost , Dual $\rightarrow$ maximize profit from resource pricing)

AI/ML Connection: Used in: (SVM , Lagrangian optimization , Deep learning constraints)

Lagrange Dual

Lagrangian :-

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i g_i(x)$$

Dual Function :-

$$\theta(\lambda) = \inf_x \mathcal{L}(x, \lambda)$$

Dual Problem :-

$$\max_{\lambda \geq 0} \theta(\lambda)$$

Key Property :- $\theta(\lambda) \leq f(x^*)$, (always gives lower bound)

Idea :- i) Fix λ , minimize over x , ii) Then maximize over λ

Real-Life Example :- Resource allocation (Assign “price” to constraints , Optimize system using prices)

AI/ML Connection :- Used in - SVM dual formulation , Convex optimization , Regularized learning

Weak Duality

Statement : For any feasible x and λ : $\theta(\lambda) \leq f(x)$

Meaning : Dual solution is always a **lower bound** of primal.

Idea : i) Dual never overestimates primal , ii) Safe approximation

Real-Life Example :- Estimated cost is always $\leq$ actual cost.

AI/ML Connection :- Ensures stability in optimization , Used in loss bounds

Strong Duality

Statement : $\min f(x) = \max \theta(\lambda)$

Condition : when - i) Problem is **convex** , ii) **Slater's condition** satisfied

Slater's Condition

Statement : If there exists x such that : $g_i(x) < 0$, (strict feasibility)

So : Strong duality holds.

Idea : i) There is a point strictly inside feasible region , ii) No boundary issues

Real-Life Example : Room with space inside (not just boundary) - We can move freely , Optimization behaves properly.

AI/ML Connection : Used in i) SVM optimality , ii) Convex ML problems , iii) Efficient training

Table

Concept	Key Idea	Meaning	Formula
Compactness	Solution exists	Finite region → solution exists	Closed + bounded
Characterization	Optimality condition	—	$\nabla f = 0$
Duality	Two problems	Two views of same problem	Primal ↔ Dual
Lagrange Dual	Transform problem	Solving via multipliers	$\max \theta(\lambda)$
Weak Duality	Lower bound	Lower bound	$\theta \leq f$
Strong Duality	Equal solutions	Exact match	$f = \theta$
Slater Condition	Feasibility	Guarantees strong duality	$g(x) < 0$

Numerical Method

1) Compactness & Existence of Solution

Problem : Minimize: $f(x) = x^2 - 4x$, over range $S = [0, 3]$

Solution

Step 1: Check compactness : Interval $[0, 3]$ is - closed , bounded . So Compact SET.

Step 2: Check continuity : $f(x)$ is continuous.

Step 3: Conclusion : By **Weierstrass theorem**: $\exists x^* \in S$

Step 4: Find minimum : $f'(x) = 2x - 4$, $2x - 4 = 0 \Rightarrow x = 2$, Checking : $f(0) = 0$, $f(2) = -4$, $f(3) = 3$

Final Answer : $x^* = 2$, $f(x^*) = -4$

2) Characterisation of Solution

Problem : Minimize : $f(x, y) = x^2 + y^2$, subject to : $x + y = 2$

Solution

Step 1: Lagrangian : $L = x^2 + y^2 + \lambda (x + y - 2)$

Step 2: Stationarity : $2x + \lambda = 0$, $2y + \lambda = 0$

Step 3: Solve : $x = y$, $x + y = 2 \Rightarrow x = 1$, $y = 1$

Final Answer : $(x^*, y^*) = (1, 1)$

3) Lagrange Dual Problem

Problem : $\min f(x) = x^2$, Subject to : $x - 1 \geq 0$

Solution

Step 1: Rewrite constraint : $g(x) = 1 - x \leq 0$

Step 2: Lagrangian : $L(x,\lambda) = x^2 + \lambda (1-x)$

Step 3: Dual function : Minimize w.r.t x : $dL/dx = 2x - \lambda = 0$, $x = \lambda/2$

Step 4: Substitute : $\theta(\lambda) = (\lambda/2)^2 + \lambda(1 - \lambda/2)$, $\theta(\lambda) = \lambda^2/4 + \lambda - \lambda^2/2 = \lambda - \lambda^2/4$

Step 5: Maximize : $d\theta/d\lambda = 1 - \lambda/2 = 0$, $\Rightarrow \lambda = 2$

Step 6: Dual value : $\theta(2) = 2 - 1 = 1$

Final Answer: Dual optimum = 1

4) Weak Duality

Problem : Given : $f(x) = x^2$, $x = 2$, Dual function: $\theta(\lambda) = \lambda - \lambda^2/4$, Take $\lambda = 1$

Solution

For : $f(2) = 4$, Dual value : $\theta(1) = 1 - 1/4 = 0.75$

Check : $\theta(\lambda) \leq f(x)$, $0.75 \leq 4$

Final Answer: Verified (Weak duality holds.)

5) Strong Duality (Slater's Condition)

Problem : Minimize: $f(x) = x^2$, Subject to: $x > 0$.

Solution

Step 1: Slater's condition : There exists x such that: $x > 0$, Example: $x = 1$

Step 2: Convexity : $f(x) = x^2$ is convex

Step 3: Conclusion : Strong duality holds

Final Answer : Dual optimum

6) Complementary Slackness

Problem : Minimize: $f(x) = x^2$, Subject to: $x - 1 \geq 0$

Solution

Step 1: Constraint : $g(x) = 1 - x \leq 0$

Step 2: Optimal solution : $x^* = 1$

Step 3: Complementary slackness : $\lambda (1 - x^*) = 0$, $\lambda(0) = 0$

Step 4: Status of λ , $\lambda = 2$

Final Answer : Condition satisfied

7) Complementary Slackness (Inactive Constraint)

Problem : Minimize: $f(x) = x^2$, Subject to : $x \geq -1$

Solution

Minimum of x^2 : $x^* = 0$

Constraint : $0 > -1 \Rightarrow$ inactive

Complementary slackness : $\lambda g(x^*) = 0$, $\lambda(-1) = 0 \Rightarrow \lambda = 0$

Final Answer : $\lambda = 0$

Practise Questions

Compactness & Characterisation (Medium & Hard)

- 1) Determine whether the set is compact: $S = \{ x \in \mathbb{R} \mid 0 < x \leq 2 \}$, Justify your answer.
- 2) Minimize: $f(x) = x^2 - 6x + 5$, over the range : $S = [0, 5]$, Check existence and find optimal solution.
- 3) Minimize: $f(x,y) = x^2 + y^2 - 2x$, subject to : $x + y = 3$, Use optimality conditions to solve.
- 4) State and verify the **first-order optimality condition** for: $f(x,y) = x^2 + y^2$, at point $(0, 0)$.
- 5) Determine whether minimum exists for: $f(x) = x^2$, over range : $S = (0,2)$, Explain using compactness.
- 6) Minimize: $f(x,y) = x^2 + y^2$, Subject to: $x^2 + y^2 \leq 4$, Identify optimal solution and justify geometrically.

Duality & Lagrange Dual (Medium & Hard)

- 7) Form the **Lagrangian dual** of: $\min x^2$, subject to : $x \geq 2$
- 8) Given: $L(x,\lambda) = x^2 + \lambda(2-x)$, Derive the dual function $\theta(\lambda)$
- 9) Find dual problem for: $\min x^2 + y^2$, subject to : $x + y = 1$
- 10) Given : $\min x^2 + y^2$, subject to : $x+y \geq 1$, Derive dual problem and solve it.
- 11) Prove that dual function: $\theta(\lambda)$, is always concave (even if primal is not convex).

Weak & Strong Duality

- 12) Verify weak duality for: $f(x) = x^2$, $x = 2$; $\theta(\lambda) = \lambda - \lambda^2/4$, for $\lambda=1$

- 13) Check whether strong duality holds for: $\min x^2$, subject to : $x > 0$
- 14) State Slater's condition and verify it for: $f(x) = x^2$, $g(x) = x-1 \leq 0$
- 15) Show that if Slater's condition holds, then: $\min f(x) = \max \theta(\lambda)$, Provide proof outline.
- 16) Give an example where: Strong duality fails , Explain why

Complementary Slackness

- 17) Check complementary slackness for: $x^* = 1$, $g(x) = 1-x \leq 0$
- 18) Given: $\lambda = 3$, $g(x^*) = 0$, Verify condition.
- 19) Solve using KKT and verify complementary slackness: $\min x^2$, subject to : $x \geq 1$
- 20) Minimize: $f(x,y) = x^2 + y^2$, subject to : $x+y \geq 2$, Solve and verify complementary slackness.